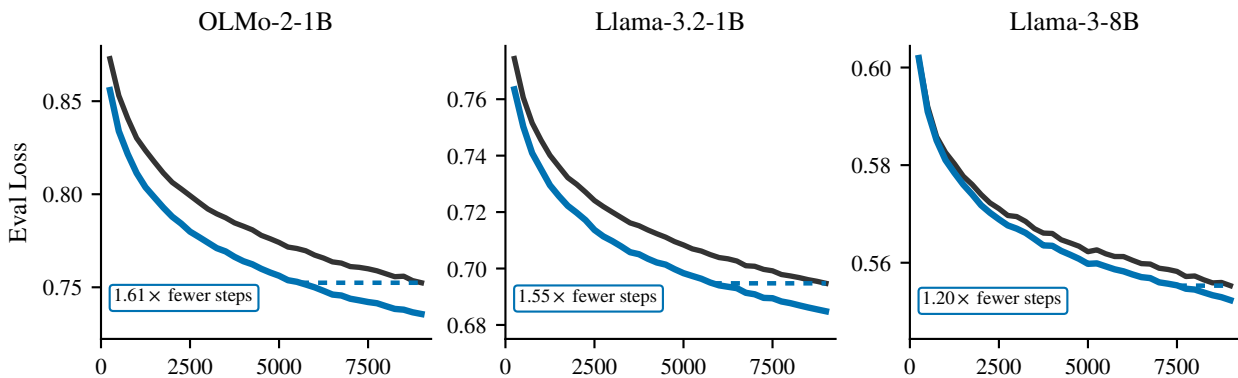
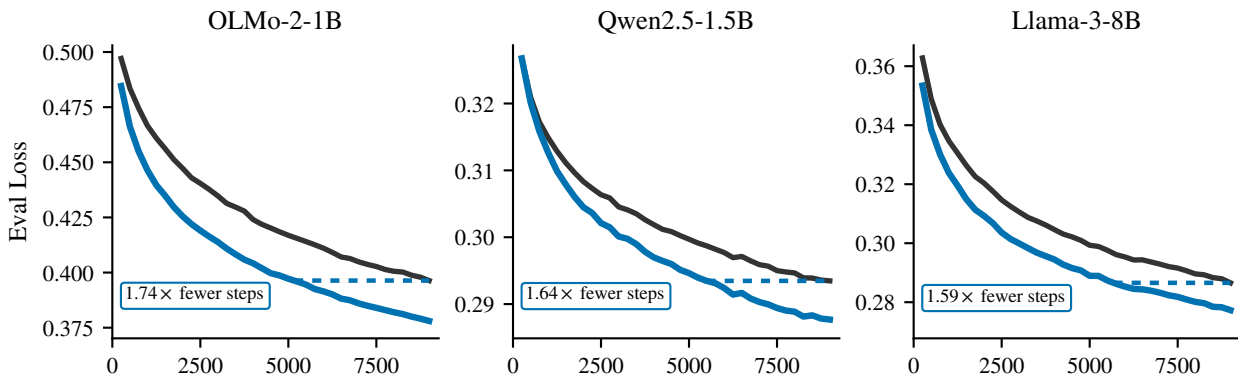


— Adam — PoLoRA

(a) Model families — code ($r=256$)



(b) Model families — math ($r=256$)



(c) Rank ladder — Llama-3.2-1B, math

